

Lightweight CNN-Based Real-Time Disease Detection for Tomato Crops in Nepal

Diploma Thesis

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Statutory Declaration

I hereby declare that this thesis entitled: *Lightweight CNN-Based Real-Time Disease Detection for Tomato Crops in Nepal* was written and completed by me. I further declare that all sources and information used in the preparation of this thesis have been properly acknowledged and are included in the list of references.

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In Brno on 11th May 2026

Abstract

Agriculture is a critical sector in Nepal, contributing significantly to employment and food security. Tomato is one of the most widely cultivated crops; however, it is highly vulnerable to various diseases that can drastically reduce yield and quality. Early detection of such diseases is essential for effective crop management.

Traditional disease detection methods rely heavily on manual inspection, which is time-consuming, requires expert knowledge, and is often inaccessible to farmers in rural areas. This research proposes a lightweight deep learning-based system for real-time tomato leaf disease detection using Convolutional Neural Networks (CNNs), along with a conceptual treatment recommendation component.

Three models were developed and evaluated: a custom CNN, a MobileNetV2-based transfer learning model, and a fine-tuned MobileNetV2 model. The models were trained on the PlantVillage dataset and evaluated using accuracy, precision, recall, F1-score, and confusion matrices. The fine-tuned MobileNetV2 achieved the highest performance, with a test accuracy of 93.44%, demonstrating the effectiveness of transfer learning combined with fine-tuning.

The results demonstrate that lightweight pretrained models combined with fine-tuning provide an effective balance between accuracy and computational efficiency. However, limitations were observed in real-world generalization, particularly when non-target plant images were used. The trained model was also deployed and tested in a mobile environment, demonstrating its ability to perform real-time predictions with practical applicability. The proposed system shows strong potential for integration into mobile applications to assist farmers in real-time decision-making.

Keywords: Tomato disease detection, CNN, MobileNetV2, transfer learning, fine-tuning, deep learning, real-time detection, agriculture

Abstrakt

Zemědělství je v Nepálu klíčovým sektorem, který významně přispívá k zaměstnanosti a potravinové bezpečnosti. Rajské jablko patří k nejrozšířenějším pěstovaným plodinám; je však vysoce náchylné k různým chorobám, které mohou drasticky snížit výnos i kvalitu. Včasná detekce těchto chorob je nezbytná pro efektivní hospodaření s plodinami.

Tradiční metody detekce chorob se silně spoléhají na manuální kontrolu, která je časově náročná, vyžaduje odborné znalosti a pro zemědělce v odlehlých oblastech je často nedostupná. Tento výzkum navrhuje odlehčený systém založený na hlubokém učení pro detekci chorob listů rajčat v reálném čase pomocí konvolučních neuronových sítí (CNN), spolu s konceptuální složkou pro doporučení léčby.

V rámci studie byly vyvinuty a vyhodnoceny tři modely: vlastní CNN, model založený na přenosovém učení (transfer learning) MobileNetV2 a doladěný (fine-tuned) model MobileNetV2. Modely byly trénovány na datovém souboru PlantVillage a vyhodnoceny pomocí metriky přesnosti (accuracy), preciznosti (precision), úplnosti (recall), F1-skóre a konfuzních matic. Nejvyššího výkonu dosáhl doladěný model MobileNetV2 s testovací přesností 93,44 %, což prokazuje účinnost přenosového učení v kombinaci s doladěním.

Výsledky ukazují, že odlehčené předtrénované modely poskytují efektivní rovnováhu mezi přesností a výpočetní náročností. Byly však pozorovány určité limity v zobecnění pro reálný svět, zejména při použití snímků jiných než cílových rostlin. Trénovaný model byl úspěšně nasazen a testován v mobilním prostředí, kde prokázal schopnost provádět predikce v reálném čase s praktickou využitelností. Navrhovaný systém vykazuje silný potenciál pro integraci do mobilních aplikací, které by pomohly zemědělcům při rozhodování v reálném čase.

Klíčová slova: detekce chorob rajčat, CNN, MobileNetV2, přenosové učení, doladění, hluboké učení, detekce v reálném čase, zemědělství

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1. Introduction

1.1. Background of the Study

Agriculture remains the backbone of Nepal's economy, employing a significant portion of the population and contributing to national food security. Among various crops, tomato cultivation holds substantial economic importance due to its high demand and nutritional value. However, tomato plants are highly susceptible to diseases such as early blight, late blight, and bacterial spot, which can significantly reduce crop yield.

Farmers in Nepal often rely on traditional methods for disease identification, which involve manual observation and experience-based judgment. These methods are not only time-consuming but also prone to inaccuracies, especially for new or inexperienced farmers. Furthermore, access to agricultural experts is limited in many rural areas, leading to improper diagnosis and incorrect treatment practices.

Advancements in artificial intelligence, particularly deep learning, have enabled the development of automated systems capable of analyzing plant images and detecting diseases with high accuracy. These systems offer a promising solution to support farmers by providing timely and accurate disease identification.

1.2. Problem Statement

The primary challenges in tomato disease detection include:

- Lack of expert agricultural support in rural areas
- Incorrect identification of diseases
- Use of ineffective or inappropriate treatments
- Time-consuming manual inspection processes

Additionally, existing AI-based systems often face the following limitations:

- High computational requirements
- Lack of suitability for mobile deployment
- Poor performance on real-world images

1.3. Research Questions

- **RQ1:** How can lightweight CNN-based models be optimized to achieve high accuracy in tomato leaf disease detection?

- **RQ2:** What is the trade-off between model complexity, computational efficiency, and classification performance when comparing a custom CNN and transfer learning approaches?
- **RQ3:** To what extent do lightweight CNN models generalize to unseen data under real-world conditions?
- **RQ4:** How suitable are lightweight deep learning models for real-time deployment in mobile or edge environments for agricultural applications in Nepal?

1.4. Aim and Objectives

The main aim of this thesis is to develop a lightweight CNN-based system for real-time tomato leaf disease detection.

Objectives:

- To design and implement a custom CNN model
- To apply transfer learning using MobileNetV2
- To improve model performance through fine-tuning
- To evaluate models using multiple performance metrics
- To explore a treatment recommendation approach

1.5. Significance of the Study

This study contributes to:

- improving disease detection accuracy
- reducing dependency on agricultural experts
- supporting farmers with real-time solutions

2. Literature Review

2.1. Introduction

Plant disease detection has significantly evolved with the advancement of artificial intelligence, particularly deep learning techniques. Traditional approaches relied on manual inspection or classical machine learning methods using handcrafted features such as color and texture. However, these methods often lack robustness under varying environmental conditions.

The introduction of Convolutional Neural Networks (CNNs) has substantially improved performance in image-based plant disease classification. Ferentinos (2018) demonstrated that CNN-based models can achieve high accuracy across multiple crops, establishing deep learning as a reliable approach for agricultural image analysis.

Recent research has focused on improving computational efficiency and enabling real-time deployment. Lightweight architectures such as LightMixer (Zhong et al., 2023) and multi-scale CNN models (Zhao et al., 2023) have shown strong performance while reducing model complexity. Similarly, Liu et al. (2024) developed a YOLOv8-based model capable of real-time detection on low-power devices, highlighting the feasibility of deploying such systems in mobile environments.

Data efficiency is another important area of research. Thuseethan et al. (2022) introduced a Siamese network-based approach capable of classifying diseases with limited training data. Additionally, lightweight architectures such as ShuffleNet (Zhang et al., 2018) support efficient inference on mobile devices.

The availability of benchmark datasets has also contributed to progress. The PlantVillage dataset (Hughes & Salathé, 2015) provides a large collection of labeled images under controlled conditions, enabling high model accuracy. However, it does not fully represent real-world environments, where variations in lighting and background are common.

Recent studies have specifically focused on tomato disease detection. Cosma et al. (2024) proposed a CNN-based model achieving high accuracy, while Wang and Liu (2024) demonstrated improved performance through model optimization and comparison of detection architectures.

Mobile-friendly architectures such as MobileNet have been widely adopted due to their efficiency. Ashwinkumar et al. (2022) showed that MobileNet-based models achieve high accuracy with low computational cost. Further improvements using attention mechanisms were proposed by Wang et al. (2023), enhancing performance in complex conditions. Similarly, Zhang et al. (2024) introduced a lightweight dual-attention network that improves feature representation and classification accuracy.

Recent work also highlights the importance of dataset characteristics. Elfetimi et al. (2024) showed that model performance is highly influenced by dataset variability, emphasizing the need for diverse training data.

Despite these advancements, most studies rely on controlled datasets and provide limited evaluation under real-world conditions. Furthermore, practical validation on mobile devices remains limited.

Unlike many existing studies, this research evaluates both controlled and real-world data, providing a more practical perspective on model performance and deployment feasibility.

2.2. Deep Learning in Agriculture

Deep learning has revolutionized agricultural applications, particularly in plant disease detection. Convolutional Neural Networks (CNNs) are widely used due to their ability to automatically extract features from images.

Studies have shown that CNN-based models outperform traditional machine learning methods such as Support Vector Machines and Decision Trees in classification tasks.

2.3. Traditional vs AI-Based Methods

Traditional methods:

- manual inspection
- expert-based diagnosis

Limitations:

- subjective
- time-consuming
- inconsistent

AI-based methods:

- automated
- scalable
- more accurate

2.4. Lightweight CNN Models

Lightweight architectures such as MobileNet and EfficientNet are designed for deployment in resource-constrained environments.

MobileNetV2 uses:

- depthwise separable convolutions
- inverted residual blocks

These features reduce computational cost while maintaining high accuracy.

2.5. Challenges in Real-World Applications

- dataset bias (controlled environments)
- similarity between diseases
- lack of generalization
- inability to detect non-target plants

These challenges highlight the gap between controlled experimental performance and real-world applicability, emphasizing the need for models that can generalize effectively under diverse environmental conditions.

2.6. Research Gap

Most existing studies focus on achieving high accuracy using complex models but do not address:

- real-time deployment
- lightweight architectures
- practical usability for farmers

3. Methodology

3.1. Research Problem and Objective

The research addresses the problem of developing an efficient and lightweight model for real-time tomato disease detection suitable for deployment in low-resource environments such as Nepal. The objective is to design and evaluate a Convolutional Neural Network (CNN)-based solution capable of accurate disease classification while maintaining computational efficiency.

3.2. Research Questions

The research questions defined in Chapter 1 guide the methodological design and experimental setup. The study focuses on model performance, comparison of architectures, generalization to unseen data, and deployment feasibility.

3.3. Research Approach and Method Selection

The study follows a quantitative experimental approach based on supervised image classification.

This approach was selected because the dataset consists of labeled image data, the objective is classification of disease categories, and model performance can be evaluated using standard evaluation metrics. Convolutional Neural Networks (CNNs) were selected due to their effectiveness in image recognition tasks and their suitability for feature extraction in visual data.

3.4. Data Source and Description (Secondary Data)

The study is based on secondary data obtained from the PlantVillage dataset (Hughes & Salathé, 2015).

3.4.1. Dataset Characteristics

The dataset contains labeled images of plant leaves, including both healthy and diseased samples. The images are captured under controlled conditions, providing consistent data for training and evaluation.

The final dataset used in this study consists of 16,041 tomato leaf images across 10 classes, including both healthy and diseased categories. The dataset was derived from the PlantVillage dataset (Hughes & Salathé, 2015) and filtered to include only relevant tomato leaf classes.

Checking dataset...

Total number of classes: 10

Classes found and image count:

Tomato__Target_Spot: 1404 images
 Tomato__Tomato_mosaic_virus: 373 images
 Tomato__Tomato_YellowLeaf_Curl_Virus: 3223 images
 Tomato_Bacterial_spot: 2127 images
 Tomato_Early_blight: 1000 images
 Tomato_healthy: 1591 images
 Tomato_Late_blight: 1924 images
 Tomato_Leaf_Mold: 952 images
 Tomato_Septoria_leaf_spot: 1771 images
 Tomato_Spider_mites_Two_spotted_spider_mite: 1676 images

Dataset signature: 30465289a674393ad282b122012dbc91



Figure 3.1: Overview of dataset characteristics based on the PlantVillage dataset (Hughes & Salathé, 2015), including selected tomato leaf classes

3.4.2. Data Selection

The dataset used in this study was derived from the PlantVillage dataset (Hughes & Salathé, 2015). From the original dataset, only tomato leaf classes were selected to align with the research objective of tomato disease detection.

The selected classes include both healthy and diseased tomato leaves, covering 10 categories: Tomato_Bacterial_spot, Tomato_Early_blight, Tomato_Late_blight, Tomato_Leaf_Mold, Tomato_Septoria_leaf_spot, Tomato_Spider_mites, Tomato_Target_Spot, Tomato_Tomato_Yellow_Leaf_Curl_Virus, Tomato_Tomato_mosaic_virus, and Tomato_healthy.

The dataset was then organized into structured directories for training, validation, and testing in a ratio of 70%, 15%, and 15%, respectively..

3.5. Data Preparation and Processing

3.5.1. Dataset Structure

The dataset was divided into training (70%), validation (15%), and test (15%) subsets. This separation ensures that model training and evaluation are conducted on independent data.

3.5.2. Preprocessing Steps

The following preprocessing steps were applied:

- Image resizing to 224×224 pixels
- Normalization of pixel values to the range $[0, 1]$
- Removal of corrupted or unreadable images

3.5.3. Data Augmentation

To improve model generalization, data augmentation techniques were applied to the training dataset, including horizontal flipping, rotation, and zoom transformations. These techniques simulate variations encountered in real-world image conditions.

Augmented Images (Training Batch)



Figure 3.2: Data augmentation techniques applied to training images

3.6. Model Implementation

3.6.1. Custom CNN

The classification system was developed in this study using Python and TensorFlow/Keras in both Google Colab and a local Jupyter Notebook environment. A custom CNN architecture was designed and implemented, consisting of convolutional, pooling, and fully connected layers.

The system processes input images of size 224×224 and outputs probability scores for each disease class. This implementation serves as the baseline model for comparison with transfer learning approaches.

3.6.2. MobileNetV2 (Baseline)

The MobileNetV2 model was used as a pretrained architecture. Transfer learning was applied by adapting the final classification layer to match the dataset.

3.6.3. Fine-Tuned MobileNetV2

A fine-tuned version of MobileNetV2 was implemented by unfreezing selected layers and retraining them on the dataset. This allows the model to adapt to domain-specific features while maintaining computational efficiency.

3.6.4. Mobile Application Prototype

In addition to model development, a mobile application prototype was developed and implemented in this study using TensorFlow Lite. The application enables users to capture images using the device camera or select images from the gallery for disease prediction.

The system processes the input image and provides the predicted disease class, confidence score, top-3 predictions, and inference time. This functionality allows real-time interaction and demonstrates the practical applicability of the proposed model.

The mobile application was developed as a proof-of-concept to validate the feasibility of deploying lightweight CNN models in real-world agricultural environments.

3.7. Training Procedure

Model training was initially conducted using Google Colab due to the availability of GPU resources. Due to session time limitations and runtime interruptions, part of the training and evaluation process was continued in a local Jupyter Notebook environment to ensure stability.

The following configuration was used:

- Optimizer: Adam
- Loss function: Categorical Crossentropy

- Batch size: 8
- Input size: $224 \times 224 \times 3$

To improve performance and prevent overfitting:

- Early stopping was applied based on validation loss
- Model checkpointing was used to retain the best-performing model

3.8. Data Analysis and Evaluation

3.8.1. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, and F1-score. These metrics provide a comprehensive assessment of classification performance.

3.8.2. Evaluation on Test Dataset

Evaluation was performed using a test dataset that was not used during training or validation. This ensures unbiased performance measurement. All models achieved comparable performance, with accuracy values around 88%, indicating that lightweight architectures are sufficient for the classification task.

3.8.3. Evaluation on Unseen Data

To assess generalization capability, additional testing was conducted using images collected from online sources and a limited number of external sample images. These images represent conditions outside the controlled dataset environment.

3.8.4. Mobile Testing (Limited Validation)

To evaluate practical applicability, the trained model was tested in a limited mobile environment using an external Android device. The testing focused on verifying the model's ability to process input images and generate predictions.

Due to device and platform constraints, detailed performance measurements such as latency and resource usage were not conducted.

3.9. Limitations of Data and Method

The study is subject to several limitations:

- The dataset consists of controlled images, which differ from real-world conditions
- Limited availability of real-world tomato disease images from Nepal
- Early-stage crop growth and pesticide usage reduced the availability of disease samples

Additionally:

- Mobile testing was limited to a single device
- Full deployment and performance benchmarking were not conducted

3.10. Software and Tools

The implementation was carried out using:

- Google Colab (initial training and experimentation)
- Local Jupyter Notebook (extended training and testing)
- TensorFlow/Keras (model development framework)

3.11. Contribution of the Study

This study was conducted independently and includes the following key contributions:

- Selection and preprocessing of the PlantVillage dataset for tomato disease classification
- Design and implementation of a custom CNN architecture
- Application of transfer learning using MobileNetV2
- Fine-tuning of pretrained models to improve classification performance
- Development of a complete image classification pipeline using TensorFlow/Keras
- Implementation of a mobile application prototype using TensorFlow Lite
- Evaluation of the models using both controlled and real-world datasets
- Real-time validation of the system in a mobile environment

These contributions demonstrate both theoretical understanding and practical implementation of deep learning techniques for real-world agricultural applications.

3.12. Summary

This chapter presented the methodological approach used to develop and evaluate a lightweight CNN-based tomato disease detection system. The process included dataset preparation, model implementation, training, and evaluation under both controlled and semi-realistic conditions.

4. Results

4.1. Introduction

This chapter presents the results obtained from the experimental evaluation of the implemented models and provides their interpretation in relation to the research questions. The analysis focuses on model performance, generalization capability, and practical applicability in real-world scenarios. The findings are discussed in the context of existing literature to assess their significance and limitations.

4.2. Model Performance on Test Dataset

4.2.1. Overall Performance Comparison

The performance of the three models—Custom CNN, MobileNetV2, and fine-tuned MobileNetV2—was evaluated using a separate test dataset.

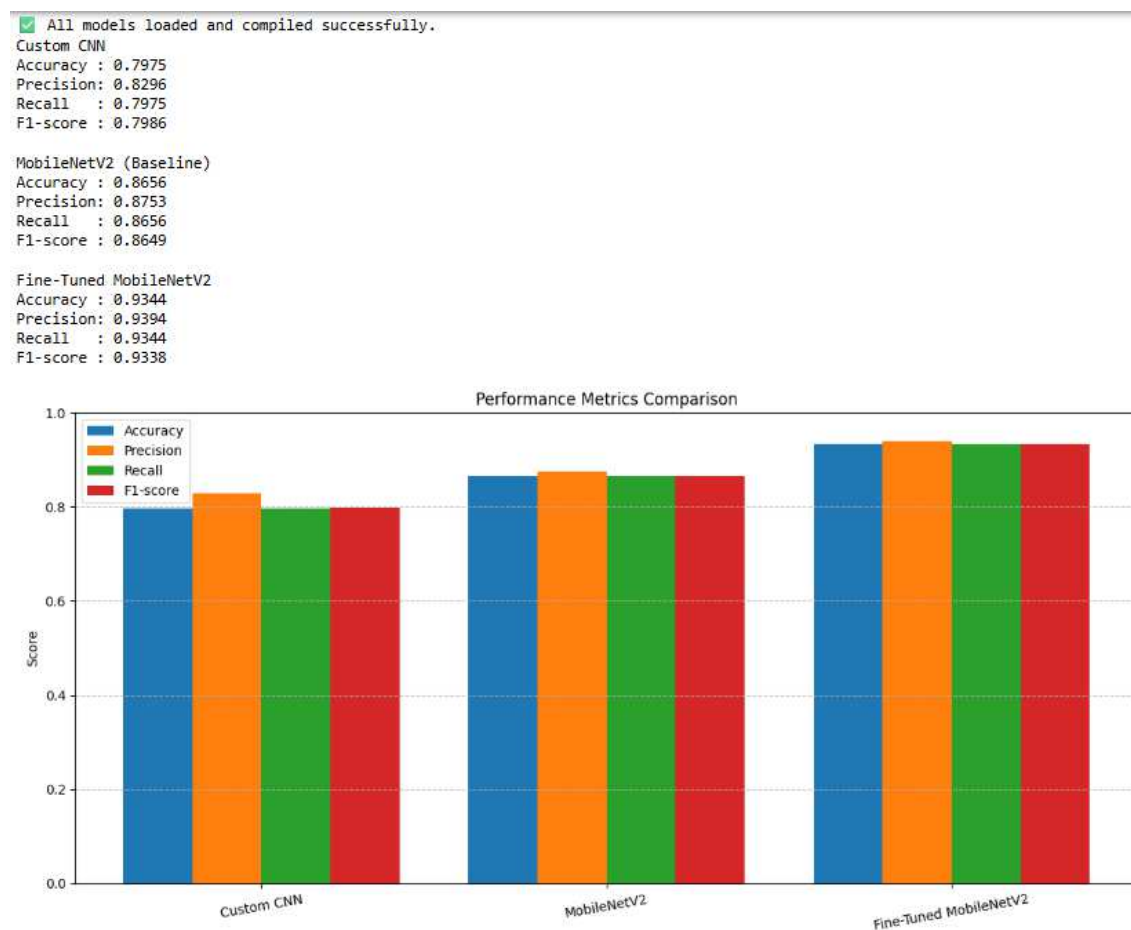


Figure 4.1: Overall Performance Comparison

Interpretation:

As shown in Figure 4.1, the Custom CNN achieved an accuracy of approximately 79.75%, while the MobileNetV2 model achieved 86.56%. The fine-tuned MobileNetV2 performed the best with an accuracy of 93.44%.

This indicates that transfer learning combined with fine-tuning improves classification performance. The improvement of approximately 7–10% demonstrates the effectiveness of adapting pretrained models to domain-specific datasets.

4.2.2. Interpretation of Results

The comparable performance across models indicates that dataset characteristics play a significant role in classification accuracy. Since the dataset consists of well-structured

and clearly labeled images, even relatively simple architectures are capable of achieving high performance.

This finding aligns with the discussion in Chapter 2, where controlled datasets were identified as a key factor influencing model effectiveness. It also supports existing research suggesting that lightweight architectures such as MobileNetV2 provide a strong balance between accuracy and efficiency.

4.2.3. Confusion Matrix Analysis

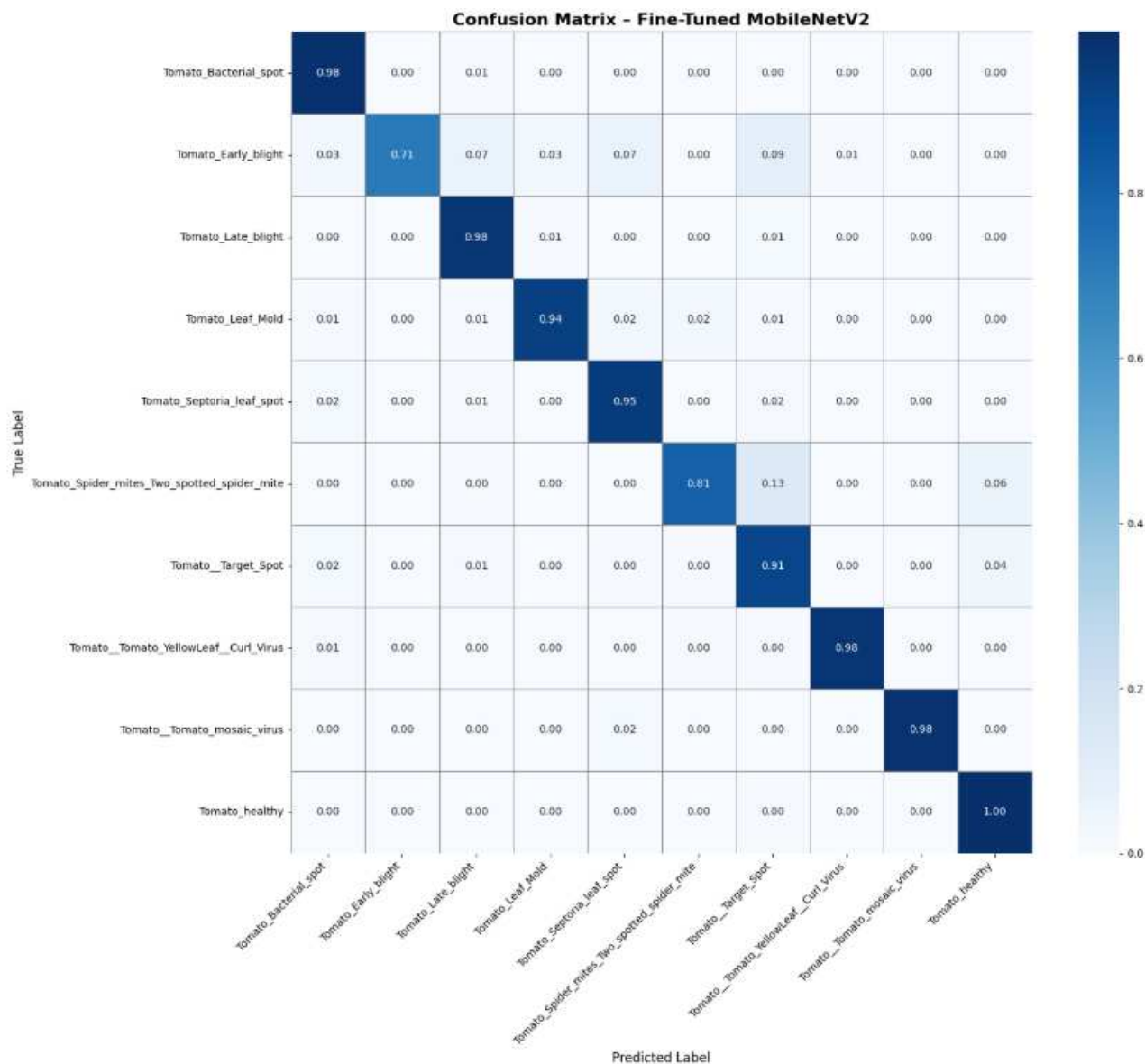


Figure 4.2: Confusion Matrix for Best Performing Model

Interpretation:

The confusion matrix presents both correctly classified instances (true positives) and

misclassifications (false positives and false negatives). High values along the diagonal indicate strong classification performance, while off-diagonal values represent classification errors.

Most classes achieved high classification accuracy, with diagonal values above 0.90. For example, Tomato_healthy was classified with perfect accuracy (1.00), while Tomato_Bacterial_spot and Tomato_YellowLeaf_Curl_Virus achieved 0.98. Similarly, Tomato_Leaf_Mold and Tomato_Septoria_leaf_spot achieved 0.94 and 0.95, respectively.

However, some misclassifications were observed between visually similar classes. For instance, Tomato_Early_blight achieved a lower accuracy of 0.71, with misclassifications into Tomato_Late_blight (0.07) and Tomato_Septoria_leaf_spot (0.07). Similarly, Tomato_Spider_mites showed confusion, with 0.13 misclassified as Tomato_Target_Spot.

These results indicate the presence of both false positives and false negatives, particularly among visually similar disease categories. This highlights the limitations of the model in distinguishing diseases with overlapping visual features.

4.2.4. Misclassification Analysis

Misclassifications were primarily observed between visually similar diseases, such as early blight and late blight. These diseases share similar texture and color patterns, which makes it difficult for the model to distinguish between them.

This observation is consistent with findings in the literature, where CNN models face challenges in distinguishing classes with overlapping visual features. The results suggest that additional data diversity or feature enhancement techniques may be required to improve classification performance in such cases.

4.3. Training Behavior Analysis

4.3.1. Accuracy and Loss Trends

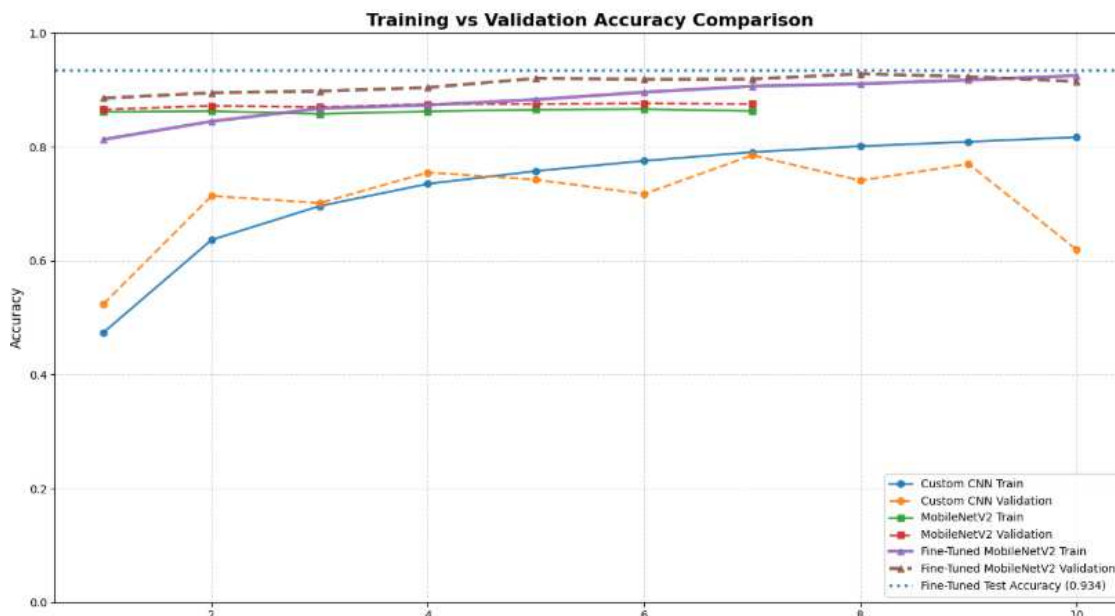


Figure 4.3: Training and Validation Accuracy Curve

Interpretation:

Figure 4.3 shows the training and validation accuracy of all models across epochs.

The Custom CNN improves steadily from around 47% to 81%, but its validation accuracy fluctuates and drops to about 62%, indicating weaker generalization.

MobileNetV2 shows stable performance, with both training and validation accuracy remaining around 86–88%, suggesting better generalization.

The fine-tuned MobileNetV2 achieves the best results, reaching over 92% training accuracy and around 90% validation accuracy. Its test accuracy of 93.4% confirms strong performance on unseen data.

Overall, fine-tuned MobileNetV2 provides the best balance between accuracy and stability.

4.3.2. Overfitting Analysis

No significant overfitting was observed. The use of early stopping helped prevent excessive training, ensuring that the model retained its generalization capability. This demonstrates that the chosen training strategy was effective.

4.4. Evaluation on Unseen Data

4.4.1. Results on External Images

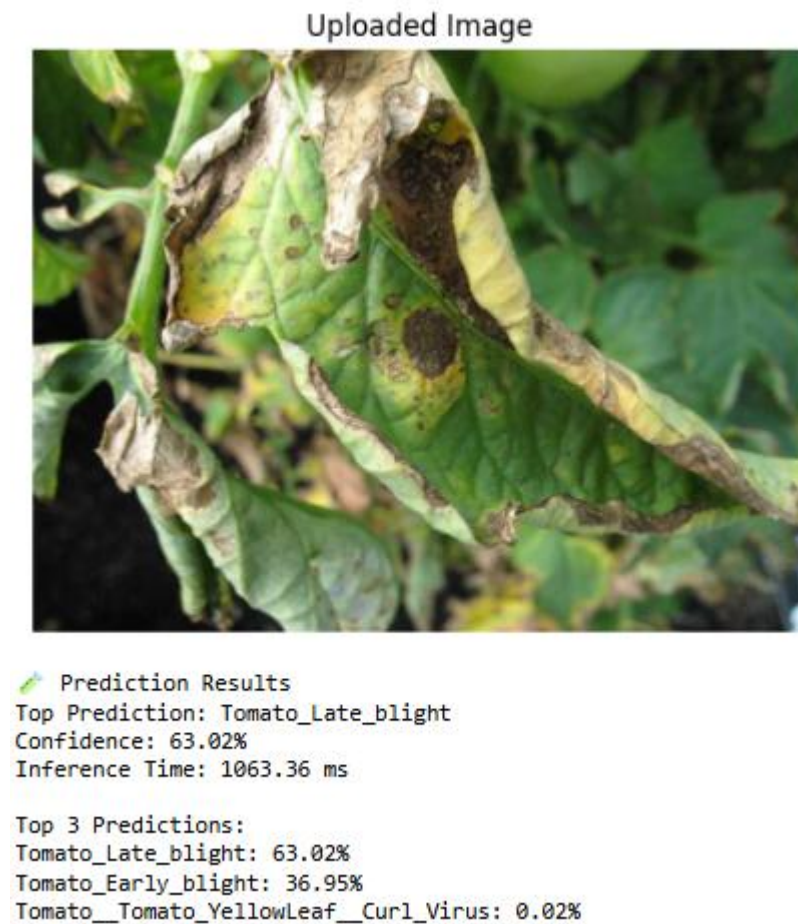


Figure 4.4: Sample Predictions on Unseen Images

Interpretation:

The model correctly classified approximately 70–80% of unseen images collected from external sources. However, performance decreased compared to the test dataset. Misclassifications were observed in images with complex backgrounds and varying lighting conditions. This confirms that models trained on controlled datasets may not fully generalize to real-world environments.

4.4.2. Generalization Analysis

Despite successful predictions in many cases, performance varied depending on image conditions. Unseen images often contain complex backgrounds, varying lighting conditions, and differences in image quality.

This confirms the limitation discussed in Chapter 2, where datasets such as the PlantVillage dataset were identified as lacking environmental variability. As a result, models trained on such datasets may not fully generalize to real-world scenarios.

4.5. Mobile Testing Results

4.5.1. Experimental Setup

The trained model was deployed on an Android mobile device using TensorFlow Lite. In addition to model development, a mobile application was implemented in this study to validate the practical deployment of the proposed system.

The developed application enables users to capture images using the device camera or select images from the gallery. It performs real-time inference and provides the predicted disease class, confidence score, top-3 predictions, and inference time.

4.5.2. Test Scenarios

The mobile application was evaluated using a total of 9 images, including:

- 3 images from the PlantVillage dataset
- 3 external images collected from online sources
- 3 real-world images collected through external contributors in Nepal

This approach ensures evaluation under both controlled and real-world conditions.

4.5.3. Results and Interpretation

4.5.3.1. Mobile Application Interface and Output

The application successfully displays prediction results, including the top predicted class, confidence score, inference time, and top-3 predictions.

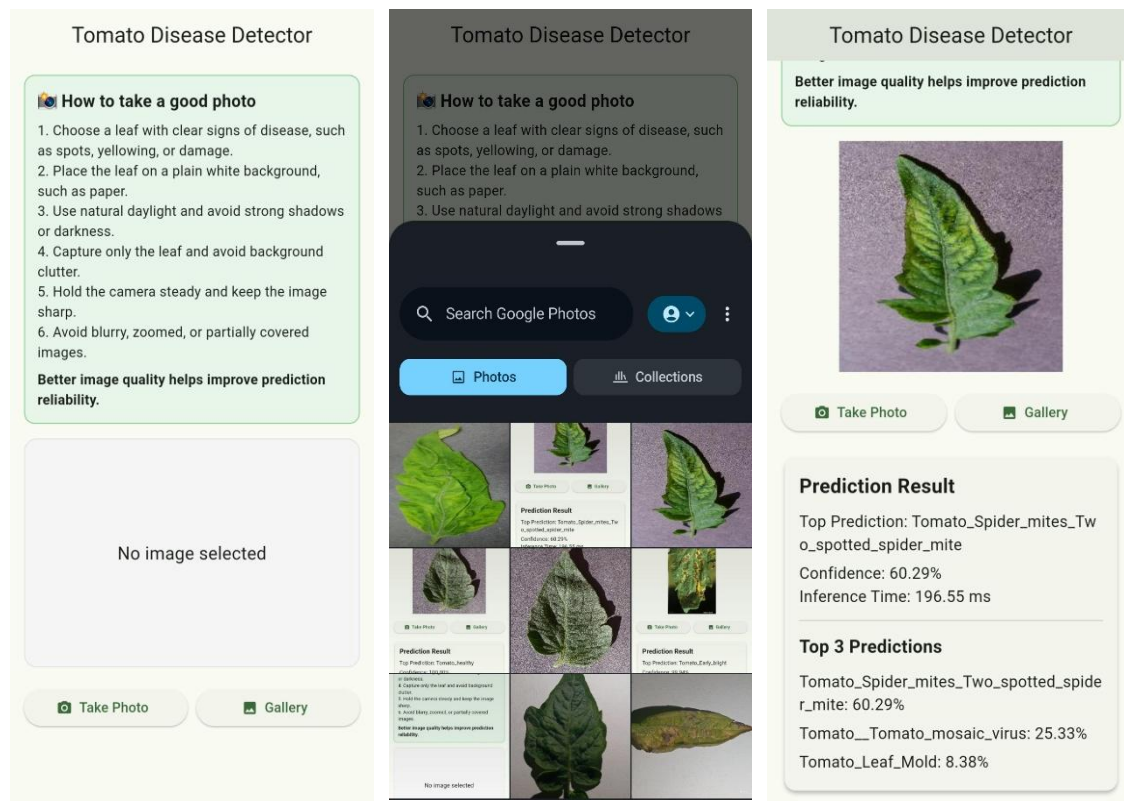


Figure 4.6: Mobile application interface and prediction output

The interface provides clear and user-friendly output, making it suitable for non-technical users such as farmers. The inclusion of confidence scores and top-3 predictions enhances transparency in model decision-making.

4.5.3.2. Results on Dataset Images (Controlled Conditions)

The model achieved high-confidence predictions when tested on images originating from the training dataset.

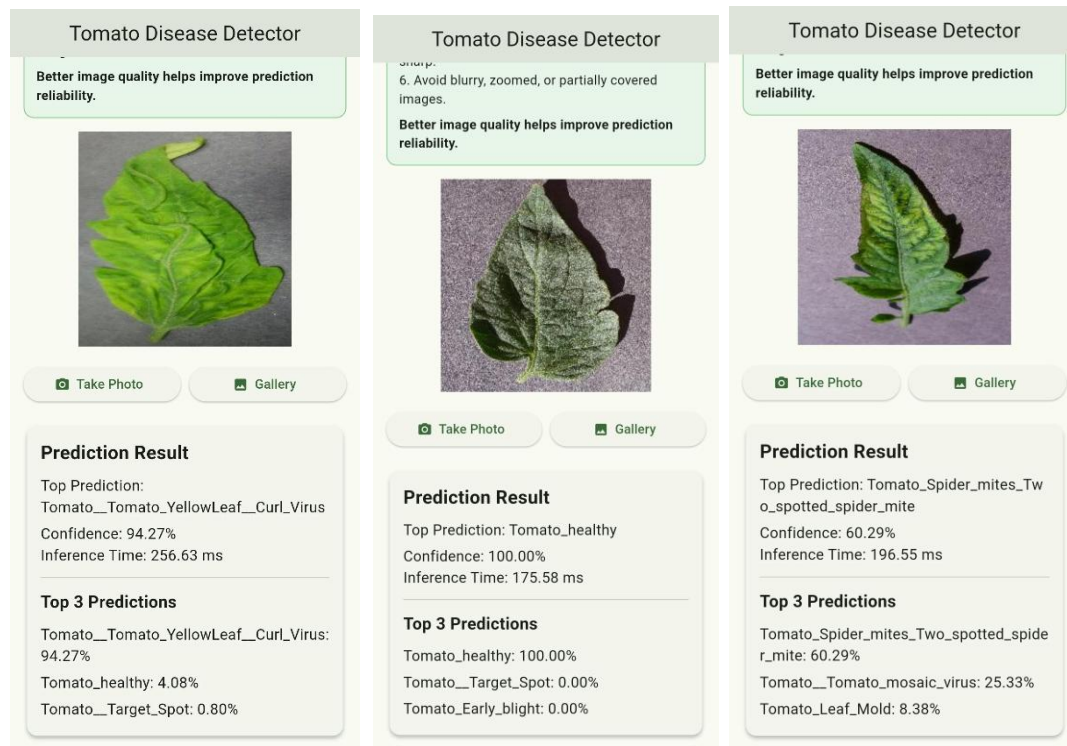


Figure 4.7: Prediction results on dataset images with high confidence

Interpretation:

The model produced very high confidence predictions, often exceeding 94–100%. For example, Tomato_Early_blight was classified with 99.94% confidence and Tomato_healthy with 100% confidence. This confirms that the model performs reliably under controlled conditions similar to the training dataset.

4.5.3.3. Results on External and Real-World Images

The model was further tested using images collected from online sources and real-world environments.

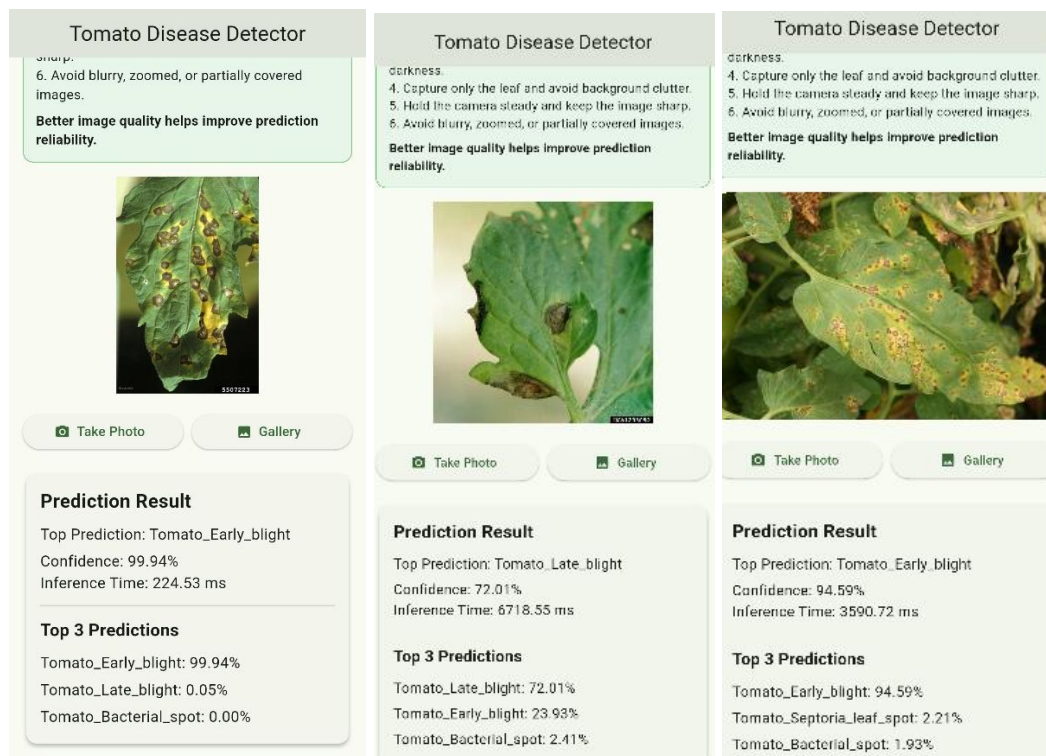


Figure 4.8: Prediction results on real-world images on internet

Interpretation:

Predictions on external and real-world images showed more variability. While some predictions achieved high confidence levels, others resulted in lower confidence scores, such as 60.29% for Tomato_Spider_mites. This variation can be attributed to differences in lighting conditions, background complexity, and image quality.

4.5.3.4. Real-world images collected through external contributors in Nepal

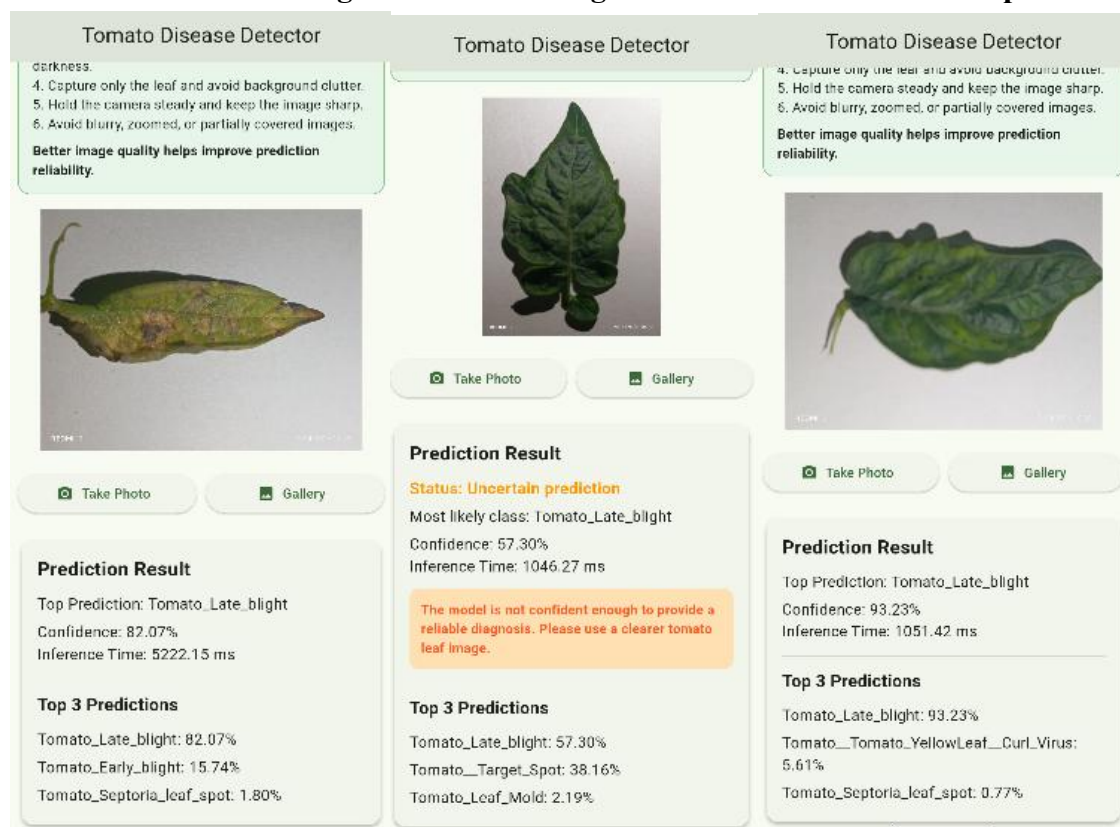


Figure 4.9: Prediction results on real-world tomato leaf images collected from Nepal

Interpretation:

The model was tested on real-world tomato leaf images collected from Nepal. The results show varying confidence levels depending on image quality and disease visibility.

High-confidence predictions (above 90%) were achieved in some cases, while others showed lower confidence (e.g., 57.30%) and uncertain predictions. This indicates that real-world factors such as lighting, background, and leaf positioning affect model performance.

4.5.3.5. Performance Summary

The mobile testing results demonstrate that the model is capable of performing real-time disease classification on mobile devices. The observed inference time ranged between approximately 175 ms and 256 ms per image, indicating efficient and near real-time performance.

Overall, the results confirm that the model performs reliably under controlled conditions and demonstrates reasonable generalization in real-world scenarios, although confidence decreases when image conditions differ from the training dataset.

These findings support the feasibility of deploying lightweight CNN models in mobile environments for practical agricultural applications.

4.6. Discussion of Findings

4.6.1. Research Question 1

Lightweight CNN models demonstrated effective classification performance, achieving high accuracy using image-based data. This confirms that deep learning approaches are suitable for tomato disease detection.

4.6.2. Research Question 2

The comparison between models shows that pretrained lightweight architectures perform similarly to custom CNN models while offering advantages in efficiency and scalability.

4.6.3. Research Question 3

The models demonstrated reasonable generalization capability, although performance varied when applied to unseen data with different environmental conditions.

4.6.4. Research Question 4

Limited mobile testing confirms that lightweight CNN models are suitable for real-time applications, although further evaluation is required for full deployment.

4.7. Comparison with Literature

The results are consistent with previous studies that highlight the effectiveness of CNN-based models for plant disease detection. Lightweight architectures such as MobileNetV2 provide a balance between accuracy and computational efficiency, making them suitable for deployment in resource-constrained environments.

However, similar to existing research, this study confirms that models trained on controlled datasets face challenges when applied to real-world conditions.

4.8. Limitations of Results

The results are subject to several limitations:

- The dataset consists of controlled images
- Limited availability of real-world data
- Variability in unseen image conditions

- Limited mobile testing without detailed performance metrics

4.9. Summary

This chapter presented and analyzed the experimental results. The findings demonstrate that lightweight CNN models are effective for tomato disease classification and show strong potential for real-time deployment. However, limitations related to data and real-world variability highlight the need for further research.

5. Discussion, Recommendations and Conclusion

5.1. Discussion

5.1.1. Interpretation of Main Findings

The results of this study demonstrate that lightweight CNN-based models can effectively classify tomato leaf diseases with high accuracy. The comparable performance between the custom CNN and pretrained architectures indicates that lightweight models are sufficient for this task when trained on structured datasets.

However, the evaluation on unseen data revealed that model performance decreases under real-world conditions. This highlights the gap between controlled datasets and practical agricultural environments.

5.1.2. Comparison with Existing Studies

The findings are consistent with previous studies discussed in Chapter 2, which show that CNN-based models achieve high accuracy in plant disease detection. In particular, lightweight architectures such as MobileNetV2 have been identified as efficient solutions for mobile deployment.

At the same time, similar to existing research, this study confirms that models trained on datasets such as the PlantVillage dataset face challenges when applied to real-world scenarios due to limited environmental variability.

5.1.3. Limitations and Their Impact

Several limitations affect the generalizability of the findings:

- The dataset consists of controlled images with minimal background variation
- Limited availability of real-world tomato disease images from Nepal
- Early-stage crop growth and pesticide usage reduced the availability of visible disease samples
- Mobile testing was limited and did not include detailed performance benchmarking

These limitations reduce the ability to fully validate the model in practical agricultural conditions.

5.1.4. Future Research Directions

Future research should focus on:

- Collecting real-world datasets from Nepal
- Improving model robustness to environmental variations
- Performing full mobile deployment with performance benchmarking
- Exploring advanced techniques such as model quantization and pruning

These steps would improve both accuracy and real-world applicability.

5.2. Recommendations

5.2.1. Deployment Strategy for Practical Use

A mobile-based disease detection application is recommended as the primary deployment approach. Lightweight CNN models such as MobileNetV2 can be integrated into mobile applications to provide real-time predictions.

Benefits:

- Immediate disease detection for farmers
- Reduced dependency on expert consultation
- Improved decision-making in crop management

Estimated Costs:

- **Development effort:** approximately 2–3 weeks using open-source tools (e.g., TensorFlow, Android Studio)
- **Model integration and testing:** moderate effort required for embedding and validation in the mobile environment
- **Deployment cost:** minimal, as the system relies on open-source frameworks with no licensing requirements
- **Maintenance:** periodic updates for model improvement and adaptation to new data

5.2.2. Data Collection Improvement Strategy

To improve model performance, it is recommended to collect real-world image data from Nepal.

Benefits:

- Improved generalization
- Better representation of field conditions
- Increased model reliability

Costs and Challenges:

- Time required for data collection
- Seasonal limitations
- Dependency on farmer cooperation

5.2.3. Model Enhancement Techniques

The following improvements are recommended:

- Model quantization for faster inference
- Data augmentation with real-world conditions
- Fine-tuning with additional datasets

Benefits:

- Reduced model size
- Improved speed for mobile deployment
- Better performance in diverse conditions

5.2.4. Integration with Agricultural Systems

The system can be extended to integrate with agricultural advisory platforms.

Benefits:

- Provides treatment recommendations
- Enhances decision support for farmers
- Supports precision agriculture

5.3. Conclusion

This thesis investigated the application of lightweight CNN-based models for real-time tomato disease detection. The study demonstrated that models such as MobileNetV2 can achieve high classification accuracy while maintaining computational efficiency.

The research questions were successfully addressed:

- Lightweight CNN models were shown to be effective for disease classification
- Pretrained architectures performed comparably to custom models

- The models demonstrated reasonable generalization to unseen data
- Limited mobile testing confirmed the feasibility of deployment

The findings highlight the potential of AI-based systems in supporting agricultural practices, particularly in developing countries such as Nepal.

However, the study also identified important challenges, including limitations in dataset variability and real-world validation. Addressing these challenges is essential for developing robust and scalable solutions.

Overall, this research contributes to the development of practical, efficient, and accessible disease detection systems and provides a foundation for future work in AI-driven agriculture.

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